

Till Tech Do Us Part: Betrayal Aversion and its Role in Algorithm Use

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Abstract

Failing to follow expert advice can have real and dangerous consequences. While any number of factors may lead a decision maker to refuse expert advice, the proliferation of algorithmic experts has further complexified the issue. One potential mechanism that restricts the acceptance of expert advice is betrayal aversion, or the strong dislike for the violation of trust norms. This study explores whether the introduction of expert algorithms in place of human experts can attenuate betrayal aversion and lead to higher overall rates of seeking expert advice. In other words, we ask: *are decision makers averse to algorithmic betrayal?* The answer to this question is uncertain ex ante. We answer this question through an experimental financial market where there is an identical risk of betrayal from either a human or algorithmic financial advisor. We find that the willingness to delegate to human experts is significantly reduced by betrayal aversion, while no betrayal aversion is exhibited towards algorithmic experts. The impact of betrayal aversion towards financial advisors is considerable: the resulting unwillingness to take the advice of the human expert leads to a 20% decrease in subsequent earnings, while no loss in earnings is observed in the algorithmic expert condition. This study has significant implications for firms, policymakers, and consumers, specifically in the financial services industry.

Key words: Algorithm Adoption, Artificial Intelligence, Behavioral Decision Making, Betrayal Aversions, Experimental Economics

JEL Codes: D91, G41, G51, C91

1. Introduction

People often seek expert advice on investing for retirement, choosing a healthcare plan, and treating illness. However, individual uptake of expert advice has consistently been found to be sub-optimal (Woodhouse and Nieusma 1997). In a number of contexts, including healthcare and financial planning, failing to adhere to expert advice can result in significant negative consequences (Seiders et al. 2015). Further increasing complexity, expert advice is rapidly being augmented by algorithms.

Consumers are faced with the decision of whether to let an algorithm manage their pension investments (Lourenço et al. 2020), suggest potential dating partners (Pahl and Van Swol 2017), or control what digital content they read and watch (Pahl and Van Swol 2017; Van Swol 2011). This trend of algorithmic advice continues to increase, and likely will for years to come (Tetlock and Gardner 2016). Importantly, in many of these instances, evidence-based forecasts made by algorithms outperform those made by their human counterparts, making this new form of expert advice even more valuable (Beck et al. 2011; Dawes 1979; Highhouse 2008).

One potential mechanism that restricts the acceptance of expert advice is what the economics literature has termed *betrayal aversion*. Betrayal aversion is defined as the strong dislike for violations of trust norms implicit in a relationship between two parties (Aimone et al. 2015). The extent of an individual's betrayal aversion mirrors the intensity of the emotional discomfort they experience in response to a betrayal, or an act of self-interested, non-reciprocated trust. While individuals may discount expert advice due to perceived flaws in the advice itself or doubts surrounding the expert's contextual understanding, these are practical concerns. Conversely, betrayal aversion operates independently of one's judgement of the advice quality or the advisor's expertise and is instead related to one's belief that the advisor will abide by the trust norms in place and relay that advice unadulterated to the individual.

Prior work has shown considerable evidence of betrayal aversion among decision-makers, even when controlling for monetary risk (Aimone et al. 2015; Aimone and Houser 2012; Aimone et al. 2014;

Bohnet et al. 2008; Bohnet and Zeckhauser 2004). One context in which betrayal aversion is often elicited is the presence of an incentive misalignment between two parties. Take, for example, an expert financial advisor whose incentive structure may encourage them to occasionally make a self-interested financial decision: by our definition, a betrayal. Upon being made aware of this incentive structure, an investor may choose to not utilize the advisor's suggested investments for a number of reasons. First, the incentive misalignment may result in lower advice quality leading to decreased financial returns or increased risk and thereby decreasing the investors total utility. In addition to those practical concerns, the investor may anticipate the emotional discomfort that they would experience knowing that upon taking the advice, the advisor may violate a trust norm implicit between them (e.g., prioritizing the investor's interests over the advisor's). This emotional discomfort, while non-monetary, also decreases the total utility realized by the investor and may lead to diminished advice adherence or an avoidance of soliciting advice at all.

Our work aims to address what changes, if any, occur to levels of exhibited betrayal aversion when an expert is replaced by an algorithm. Recent work has highlighted the practical consequences of incentive misalignments in algorithms (Lee and Hosanagar 2019; Levy 2021; Liu et al. 2023). However, to our knowledge, no work has examined the emotional discomfort elicited in conjunction. While intuition may imply that an algorithm's lack of agency and intentionality diminishes its ability to truly *betray*, many accounts of betrayal aversion bypass the need for human intentions completely and introduce a significant potential for betrayal aversion to persist for non-human advisors. For example, Koehler and Gershoff (2003) show that safety products such as airbags or vaccines can elicit feelings of betrayal in their users. Other work shows that humans mindlessly attribute social rules and expectations to computers, even when they know the computers lack intentionality (Nass and Moon 2000; Nass et al. 1994). However, no work has examined if betrayal aversion persists when a human expert is augmented or replaced by an algorithm. This disconnect is essential to understanding the role of algorithms in motivating the acceptance of expert advice. Therefore, we ask: *are humans averse to algorithmic betrayal?*

Given the rapid growth of algorithm use in financial market trading¹ and its potential to increase efficiency, transparency, and capacity in the trading process (Bell and Gana 2012) , we contextualize our work within the financial services industry specifically. To answer our central research question, we conduct an economic lab experiment in which participants play a 40-round financial trading game using a simulated market structure and a real financial trading algorithm designed to give advice that would be comparable to what consumers would receive from both human and algorithmic experts in a real-life setting. Participants are given the choice in each round to either make their own decision on how much of their endowment to use to purchase a risky asset or use the advice of an expert. No deception was used in our experiment and all participants were compensated based on their actual investment decisions.

The experiment employs a two factor (2x3) between-subjects design. The first manipulated factor is the *presentation* of the expert as either human or algorithm; importantly, the underlying algorithm that generates the advice is identical for both the human and the algorithm expert. The second manipulated factor is the presence of a salient betrayal risk where treated participants are informed that there is a slight chance that the human or algorithmic expert may make a self-interested decision on their behalf. To isolate the emotional response to the betrayal (betrayal aversion), we include an error condition where participants learn that there is a slight chance of an accidental error occurring. Ex ante, one would expect the error condition to elicit similar impacts on the perception of the monetary performance of the expert, while the lack of intentionality would limit the presence of betrayal aversion. We recruited 275 participants from Amazon Mechanical Turk (mTurk) to take part in the experiment over a video sharing platform in, what we believe to be, one of the first digitally-face-to-face economic experiments conducted on mTurk.

Our analysis captures changes in advisor usage trends in the betrayal risk (both human and algorithm), error risk, and no disclaimer (control) groups. When in a human advisor condition, we find a

¹ <https://therobusttrader.com/what-percentage-of-trading-is-algorithmic/>

roughly 16% decrease ($\beta=-0.159$, $p=0.024$) in advisor usage when informed of a betrayal risk but no decrease when only informed of a risk of accidental error ($\beta=0.007$, $p=0.924$). This result implies that the effect we measure is due to betrayal aversion and not a perceived decrease in expected monetary return. Interestingly, we do not find this effect in the algorithm conditions ($\beta=0.055$, $p=0.431$). These findings imply that substituting an algorithmic advisor for a human advisor can significantly attenuate betrayal aversion. We also observe that betrayal aversion results in a significant economic loss, resulting in a loss of \$2.15 ($p=0.072$), an almost 20% decrease in overall returns for affected participants. Analysis of exit questions substantiates that betrayal aversion may be a key aspect of this difference: participants in the betrayal condition with a human advisor reported feeling more concerned about being misled ($p = 0.006$), having their trust violated ($p = 0.010$), and most importantly, feeling betrayed ($p = 0.001$). We do not find these effects when the advisor was an algorithm, nor in the error-risk condition.

We find similar effects in a second experiment that utilizes the same experimental design while 1) drawing from a different population, undergraduate students recruited from an undergraduate economics lab and 2) utilizing a different human advisor. We find that risk of betrayal decreased uptake of the human advisor by 12% ($p = 0.0901$) and earnings by \$2.13 ($p = 0.003$). The effects were again attenuated when assigned to an algorithm advisor ($p = 0.026$). Although the smaller sample size resulted in the effect of human betrayal aversion on advisor utilization to be significant only at the 10% level, these results generally confirm the findings from the main experiment while varying the research platform, participant sample, and advisor selection. Note that the lab utilized for this study explicitly prohibits any form of deception in experimental procedures, so this experiment reduces concerns about the credibility of the information provided to participants when compared to the mTurk sample.

Our work contributes to the nascent literature exploring the acceptance, or lack thereof, of expert algorithms. The majority of the literature (see Logg et al. (2019)) for a notable exception) highlights decision-maker's preference towards human experts (Dietvorst et al. 2015; Longoni et al. 2019) and identifies traits that algorithms *lack* that may lead an individual to exhibit "algorithm aversion" (Castelo

et al. 2019). Proposed solutions to the nonacceptance of algorithmic experts have focused on either enhancing the human involvement in the algorithmic decision making by allowing users to add feedback (Dietvorst et al. 2018), making an algorithm more human-like by highlighting the affective abilities of the tool (Castelo et al. 2019) or anthropomorphizing the algorithms (Schanke et al. 2021). Our results highlight an important nuance to the conclusions of prior work: reducing the human element associated with algorithmic experts can remove some of the traditional barriers to the acceptance of expert advice.

We also contribute to this literature by identifying a potential strength of algorithmic experts and introducing a novel phenomenon to the algorithmic adoption literature – betrayal aversion. This phenomenon, largely unstudied by the information systems discipline, has been shown by behavioral and experimental economists to significantly influence an individual’s decision-making. Given that the economics literature has shown the phenomenon to exist even when the betrayal comes from inanimate products or objects, its role in the human-algorithm relationship is essential to explore. However, no work, in economics or information systems, has explored the extent to which betrayal aversion may persist (or not) when a human is replaced by an algorithm. By bridging these two literatures, we show that despite prior evidence showing the potential of inanimate objects to elicit betrayal aversion, algorithms may attenuate the phenomenon instead.

We also provide valuable insights to the Financial Technology (FinTech) literature. Algorithmic trading services have grown substantially and trading services augmented by artificial intelligence are likely in the near future (Gomber et al. 2018). Recent estimates show that between 60 and 75% of the total trading volume in the U.S. involves some form of algorithmic trading, but more research is needed to understand the implications of this trend (Alt et al. 2018; Cao et al. 2020; Gomber et al. 2018; Hendershott et al. 2021; Kou 2019). We specifically respond to Hendershott et al. (2021)) by examining the role of algorithmic financial advice and providing a potential solution to a significant barrier to the acceptance of expert advice within financial services.

2. Conceptual Background: Algorithms and Betrayal

Our work examines two primary areas of the literature. First, we look to previous work on algorithm adoption and aversion that focuses on specific drivers of use and disuse to better understand predictors of algorithmic acceptance. Then, we explore the betrayal aversion literature in economics and introduce it as a previously unexamined factor in algorithm uptake. In the section that follows, we contextualize our work to financial technology and examine the theoretical expectations of the effect of betrayal aversion on the use of trading algorithms.

2.1 Algorithm Adoption and Aversion

Prior work considering the use and adoption of information technology has strived to consider user's social and emotional beliefs (Benbasat and Wang 2005; Qiu and Benbasat 2005; Venkatesh and Davis 2000). This has led to the perception of IT artifacts as “social actors” (Al-Natour and Benbasat 2009; Reeves and Nass 1996). In the Computers are Social Actors (CASA) paradigm, humans have the same social rules and expectations of technology that they have of other humans (Nass et al. 1994). This paradigm now has renewed importance as digital ecosystems built on powerful algorithms are influencing the daily lives of their users (Parker et al. 2017). Artificial intelligence and algorithmic decision makers are augmenting or replacing their human counterparts, even in domains like medicine (Jussupow et al. 2021), and online services are increasingly using algorithms to determine the content users see (Orlikowski and Scott 2015). Given the increased prevalence of algorithm-run services and platforms (Ransbotham et al. 2018), research that considers the complexities of algorithmic use and adoption is of growing importance.

While algorithm adoption has grown, some research has shown that overall, people still prefer human decision makers over algorithmic decision makers (Diab et al. 2011; Eastwood et al. 2012). Dietvorst et al. (2015)) terms this *algorithm aversion*, when individuals avoid using algorithms, after they see them make an error (See Burton et al. (2020)) for a review).

A number of potential drivers to this phenomenon have been introduced. First, false expectations lead to an individual's unwillingness to use and accept an algorithm. This may occur because individuals believe that human error is random whereas algorithmic error demonstrates that the entire *system* is flawed instead of a one-time error (Dietvorst et al. 2015; Dietvorst et al. 2018; Highhouse 2008). Alternatively, individuals may value forming a professional relationship with those from whom they seek advice (Alexander et al. 2018; Önköl et al. 2009; Prah and Van Swol 2017). Further, Castelo et al. (2019)) finds that algorithm aversion is dependent on task type, for example, when a task is perceived as subjective, individuals are less inclined to trust an algorithm. More recently, work has highlighted the role of incentive misalignment in driving a lack of algorithmic adoption (Lee and Hosanagar 2019; Levy 2021; Liu et al. 2023), focusing on more practical considerations while leaving the emotional response relatively unexplored.

However, when individuals perceive that an algorithm is able to learn from experience, the effect of algorithm aversion is lessened (Berger et al. 2021). Additionally, when an individual gains experience with an algorithm, and receives feedback outlining the superior performance of the algorithm, aversion decreases over time (Filiz et al. 2021). Research also shows that human decision makers fear relinquishing control, thus preferring a certain level of power over an algorithm's advice (Colarelli and Thompson 2008; Scherer et al. 2015). Dietvorst et al. (2018)) further explores the integration of human-in-the-loop decision making where the human decision maker oversees the algorithm's processes, creating a perception of control.

There is also a small but growing literature considering the opposite effect: that people prefer algorithms to humans. Some work has shown that when individuals are asked to remember important information, they opt to outsource the task to algorithms, showing a preference towards algorithms over their own abilities (Sparrow et al. 2011). Further, Logg et al. (2019)) shows that individuals rely more heavily on algorithmic advice than advice from others or their own judgement, a phenomenon they term "*algorithm appreciation*." While most of the past work has proposed solutions to this debate by focusing

on making algorithms more *human* (Schanke et al. 2021; Wilson et al. 2017), no work, to our knowledge, has considered the potential for algorithms to bypass traditional barriers to the willingness to seek advice often faced by human experts. We contribute to this literature by using insights from behavioral economics to explore the role of betrayal aversion in algorithm adoption.

2.2 Betrayal Aversion

Betrayal aversion is defined as the strong dislike for violations of trust norms implicit in a relationship between two parties (Aimone et al. 2015). Early explorations of betrayal aversion focused on the relationship between a perceived “duty to protect” and the extent of betrayal aversion. For example, Koehler and Gershoff (2003) show that participants view a betrayal as more severe when a member of the military commits treason than when its committed by an individual in an unrelated career. The same phenomenon was shown in criminal punishment where, for example, people believe that day care workers who abuse a child should be punished more harshly than a janitor who abuses a child in the same way. This has led to demands to revise sentencing guidelines to make them sensitive to the extent to which the defendant’s role included ensuring an individual’s wellbeing (Shnoor 2009).

Later work expanded on these ideas, focusing on removing elements of risk and trust to isolate betrayal aversion in traditional economic trust games. The general method of studying betrayal aversion through this lens involves having one group of participants play a trust game with random chance determining the player’s financial return, instead of another human player. This work shows that participants require a higher minimum acceptable probability (MAP) of earning a return in a trust game played with a human compared to a similar game involving random chance, providing evidence of non-monetary betrayal aversion (Bohnet et al. 2008; Bohnet et al. 2010; Bohnet and Zeckhauser 2004; Hong and Bohnet 2007). This phenomenon has expanded beyond economic decision making to other fields as well. Some work has shown that managers are influenced by betrayal aversion when developing relationships with their employees (Birnberg and Zhang 2010). The results showed that some managers

spend more money to prevent betrayal than they could have conceivably lost from the betrayal itself, implying that betrayal *aversion*, not just the betrayal, may lead to a decrease in expected monetary return.

While the majority of the literature has focused on human-to-human interactions, some work has explored the potential for non-human agents to elicit fears of betrayal. Koehler and Gershoff (2003) show this phenomenon through the elicitation of opinions on different consumer products. They find that individuals often prefer less effective products to products that have a small chance of causing the harm they were designed to protect against. For example, when given the option between an airbag with a 2% chance of driver death in a serious accident or one with a 1% chance of death plus a 0.01% chance of the airbag causing the death when the driver would have otherwise survived, participants strongly prefer the former. Other work has shown that consumers who feel betrayed by products punish the firm by way of complaints or negative reviews (Grégoire and Fisher 2008).

Firms are increasingly faced with the choice to adopt algorithmic tools and services into their underlying business models and individuals are subsequently tasked with evaluating and deciding to use those tools. Given this trend, the presence of betrayal aversion in both consumers and employees has the potential to influence levels of algorithmic adoption and use.

3. Theoretical Development

Betrayal aversion with respect to algorithms has not yet been explored. To lend concreteness to our theoretical expectations, we focus on algorithm aversion in the financial services context. As mentioned previously, algorithmic trading (AT) has grown substantially in financial markets around the world. The use of AT has the ability to increase efficiency, transparency, and capacity in the trading process (Bell and Gana 2012). These increases are realized through the rapid collection of public information (Chakrabarty et al. 2015) and the ease in converting that information into prices (Brogaard et al. 2014). Additionally, AT can enhance liquidity and increase the informativeness of price quotes (Hendershott et al. 2011).

Although investors and traders have increasingly developed trading algorithms designed to mimic the trends and actions of human traders (Hendershott et al. 2021), consumer unwillingness to adopt often superior algorithms can result in lower returns (Ge et al. 2021). In addition, if betrayal aversion persists in this context, the phenomenon should be considered in the design and marketing of algorithmic tools. While we defer to future work for specific design recommendations, we highlight for the first time the potential of algorithmic betrayal aversion and open the door for further explorations. In the rest of this section, we use the context of trading algorithms to explore the role of betrayal aversion in algorithmic use and leverage common assumptions of utility maximizing agents (e.g. Bernard et al. (2015))). What follows is not a formal analytical model but rather a conceptual exercise to help organize competing dynamics and articulate ex ante expectations around algorithmic betrayal aversion.

Assume that a decision maker has a basic utility function such that:

$$U(E, \sigma) = E - \frac{1}{2} R \sigma^2 \quad (1)$$

where E is the expected return or mean outcome of a financial market trade that follows a well-defined probability distribution function with a variance of σ^2 . R is a constant risk aversion parameter that reflects the decision maker's risk preference (reflective of the Arrow-Pratt risk aversion measure first introduced by Pratt (1964)) and Arrow (1971))). Assume that the decision maker has a choice between two financial strategies for the trading decision. They can choose to make their own decision (S), or they can choose to use the recommendation of a financial advisor (A) such that:

$$U(E_i, \sigma_i) = E_i - \frac{1}{2} R \sigma_i^2, \text{ where } i \in [A, S] \quad (2)$$

Before making the choice, they are told the following:

$$E_A > E_S \quad (3)$$

Therefore, an individual would choose to use the advisor's recommendation over their own decision if the increase in expected return is higher than the difference in the variance scaled by the risk aversion parameter such that:

$$E_A - E_S > \frac{1}{2}R(\sigma_A^2 - \sigma_S^2) \quad (4)$$

However, they are also told that if A is chosen, there is a small probability of event B, or a betrayal, also occurring. This probability can be expressed as E_B (the mean probability of a betrayal occurring). The betrayal occurs when the advisor makes a self-interested choice that may or may not align with the ideal trading strategy of the decision maker, with equal probability. In other words, the betrayal may increase or decrease the investor's return. However, it is classified as a betrayal because the advisor makes the choice that best suits their own interests with no regards for the investor. If the goals of the advisor and the investor happen to align when a betrayal occurs, the investor may still financially benefit².

While the chance of a betrayal does not decrease the expected monetary return such that $E_{A|B=1} = E_{A|B=0}$, it would necessarily increase the variance. Let us denote $\sigma_{A+}^2 = (\sigma_A^2 | E_B > 0)$ and assume that $\sigma_{A+}^2 > \sigma_A^2$. Given the extensive empirical findings described in the previous section, we also posit that the utility function also includes a non-monetary element β , or betrayal aversion, in which an individual exhibits an emotional cost from a betrayal that is distinct from any consideration of expected monetary returns. The difference in utility of choosing A over S would then be:

$$\Delta U = \left[E_A - \frac{1}{2}R\sigma_{A+}^2 - \beta E_{B|A} \right] - \left[E_S - \frac{1}{2}R\sigma_S^2 - \beta E_{B|S} \right] \quad (5)$$

where β represents the level of betrayal aversion the individual exhibits and E_B is the expected betrayal outcome $[0,1]$ as described above. We assume that $E_{B|S} = 0$ implying that a betrayal could not occur when choosing to invest themselves.

Let us assume that there is an individual with a unique risk aversion parameter, R^* , who, absent any risk of betrayal, is indifferent between the two trading strategies assuming $E_A > E_S$ and $\sigma_A^2 > \sigma_S^2$:

² While a real-world betrayal may come with financial consequences, for the sake of simplicity, we assume here that the expected return is constant. This allows us to isolate the betrayal aversion parameter more easily, increasing the clarity of our expectations.

$$E_A - \frac{1}{2} R^* \sigma_A^2 = E_S - \frac{1}{2} R^* \sigma_S^2 \quad (6)$$

$$R^* = \frac{2[E_A - E_S]}{\sigma_A^2 - \sigma_S^2} \quad (7)$$

If this individual also exhibits some level of betrayal aversion and the probability of a betrayal is non-zero, such that $\beta > 0$; $E_{B|A} > 0$; $\sigma_{A+}^2 > \sigma_A^2$, they will choose to make their own trading decision over taking the advisor's, as shown below:

$$E_A - \frac{1}{2} R^* \sigma_{A+}^2 - \beta E_{B|A} < E_S - \frac{1}{2} R^* \sigma_S^2 - \beta E_{B|S} \quad (8)$$

$$-\beta E_{B|A} < (E_S - E_A) + \frac{1}{2} \frac{2[E_A - E_S]}{(\sigma_{A+}^2 - \sigma_S^2)} (\sigma_{A+}^2 - \sigma_S^2) \quad (9)$$

$$\beta E_{B|A} > 0 \quad (10)$$

Several prior papers that attempt to isolate betrayal aversion in this way (see Section 2.2), found that participants engaging in games in which their payoff was determined by a human required a higher minimum acceptable probability of earning a positive return than participants whose payoff was determined by random chance (Bohnet et al. 2008; Bohnet et al. 2010; Bohnet and Zeckhauser 2004; Hong and Bohnet 2007). This work provided evidence for the existence of a non-monetary, emotional disutility that arose from the risk of being betrayed. Therefore, in line with these findings, we hypothesize:

H1: Holding the objective monetary return constant, the existence of betrayal aversion will decrease the willingness to outsource a decision to a human expert when there is a chance of betrayal.

This hypothesis is consistent with previous research on betrayal aversion towards a human counterpart. Since previous work has not examined betrayal aversion within financial investing, we first aim to explore whether betrayal aversion exists in this context. We then examine whether levels of betrayal aversion differ when a human advisor is replaced with an algorithmic advisor. As stated in earlier sections, our formulation of betrayal aversion is “a strong dislike for the violation of trust norms

implicit in a relationship between two parties.” To establish *ex ante* expectations for the strength of betrayal aversion for algorithms compared to humans, we can examine the hypothetical strength of trust norms implicit between an individual and an algorithm. Prior work has examined individual’s expectations of algorithms or non-human agents and have found that assessments of fairness are likely to differ between algorithms and humans. For example, Helberger et al. (2020) find that when individuals determine if a decision maker is “fair” they utilize a more procedural definition of fairness when the decision maker is an algorithm instead of a human.

Work that examines individual’s “mind perception”, or the ability to ascribe mental capacities to other entities (Gray et al. 2012), find that social robots are viewed as having limited moral agency and responsibility compared to human actors. Interestingly, despite ranking very low on *experience*, the capacity to feel and sense, individuals viewed the robot as having slightly more *agency* than a chimpanzee and only slightly less agency than a child. Agency attributions were also highly correlated to deserving punishment for wrongdoing. Given the inherent link between agency, moral judgement, and responsibility, this leaves open the possibility that algorithms may still elicit some level of betrayal aversion from those it interacts it. This aligns with prior work that has shown that, in some instances, humans mindlessly apply social rules and expectations to computers, even when they know that the computers lack feelings and intentionality (Nass and Moon 2000; Nass et al. 1994).

Therefore, we anticipate betrayal aversion persisting for algorithmic experts. In other words, our expectations imply that:

$$\beta_{Algorithm} > 0 \quad (11)$$

leading us to hypothesize:

H2: Holding the objective monetary return constant, the willingness to outsource a decision to an algorithmic expert will decrease when there is a chance of betrayal.

However, the work described above, while highlighting that some agency was attributed to robots, showed far lower agency attributions towards the robot than towards human adults. Further, Gray and Wegner (2012) find that humans have a deep-seated expectation that technological actors (e.g., robots, chatbots) lack in *experience*, the capacity to feel and sense, and agency. Gray et al. (2012) expand on these findings and highlight that the application of moral judgment (labeling actors as good or bad) is fundamentally intertwined with dimensions of mind perception, which algorithms inherently lack. Relatedly, individuals cast less moral judgement on a drug company that increased the price of an essential medication through an intermediate company than when taking the same action directly (Paharia et al. 2009). In other words, humans view moral violations less severely when the harm is carried out indirectly (Greene and Sinnott-Armstrong 2007; Hauser 2006). This leaves open the possibility that organizations that utilize algorithms over human actors may be held less accountable for potential harm caused by them.

The presence of betrayal aversion is fundamentally tied to this same perception of moral judgement. This is best highlighted by work in neuroscience showing that the same area of the brain (anterior insula) is activated both when individuals are exhibiting betrayal aversion (Aimone et al. 2014) and when assigning moral judgement (Moll et al. 2005; Ying et al. 2018) and identifying social norm violations (Cheng et al. 2017; Xiang et al. 2013).

Overall, extensive work shows that humans perceive algorithms to lack in the primary dimensions of mind perception. These dimensions serve as the “essence” of moral judgement (Gray et al. 2012) and the identification of social norm violations. Betrayal aversion, defined through this same norm violation lens, is therefore likely to differ as the ability to assign agency and experience decrease. In other words, we expect that levels of betrayal aversion will decrease when human actors are replaced by algorithms implying that:

$$\beta_{Human} > \beta_{Algorithm} \quad (12)$$

Therefore, we hypothesize:

H3: Holding the objective monetary return constant, there will be a smaller decrease in the willingness to outsource a decision to an algorithm compared to a human when there is a chance of betrayal.

4. Financial Investment Game

4.1 Experimental Design

Our experiment utilized an online financial investment game developed through oTree Version 5.9 (Chen et al. 2016), a Python framework commonly used in experimental economics, and hosted on a Heroku web server to allow for remote browser-based access to the game. The player's performance in the game determined their true financial payout after the game was completed.

Game Design: The game consisted of 40 rounds of investing in a simulated financial market that had a single risky asset and two market states: good or bad. The choice was made to utilize a high number of rounds in the game in order to increase the likelihood that participants would experience various fluctuations in the state of the market. At the beginning of the game, the player was given a set endowment of research points. In each round, the player decided how many of those points they would invest in the asset. The decision process was intended to replicate a simple market transaction in which they decided what proportion of their endowment would be invested in the asset, the price of the asset moved, and following the price movement, their stake in the asset was sold and their earnings or losses were added back to their endowment.

The state of the market varied across rounds. In a good market state, the risky asset had a higher probability of increasing in price and vice versa for a bad market state. The market state was partially dependent on the state of the market in the prior round with a state-switch rate of 35%. For example, if the immediately preceding round's market was in a bad state, the current

round's market had a 65% of also being in a bad state. In every round, the game's backend tracked the changes in the asset price and used Bayesian probability rules to predict the state of the market and derive a recommended investment for the present round. We pilot tested the game's recommendation system and found that, on average, it increased participant returns by over 60%, making it an effective tool for investing in our simulated market. For a detailed mathematical explanation of how the market and the recommendation system operated, see Appendix A.

In every round, the player first made the choice to use the game's recommended investment or to make a decision entirely on their own. The recommended investment was not shown prior to the decision being made and the player only learned the investment amount after the individual round was completed and the returns were distributed back to the player. If a player had chosen to invest on their own, they were directed to the investment page (see Figure 1) where they were shown their current endowment and asked to decide how much of that endowment they would invest in the asset. Otherwise, if the player had chosen to use the recommended investment, the investment page was bypassed and they were immediately shown the results of that round's investment. The results page (see Figure 2) displayed how much they had invested, the percentage change that the asset's price had moved, the current round's return and cumulative returns across all prior rounds, and two graphs: one that tracked the price movement of the asset and one that tracked their total endowment. These graphs were intended to increase the realism of the game and allowed participants to compare their own performance to the overall price movement of the asset.

Your Investment Choice

You currently have 1000.0 research points. What percent of these points would you like to invest in the asset for Round 1?

50%

This will invest 500 of your points in the asset.

[Next](#)

Figure 1: Investment Page

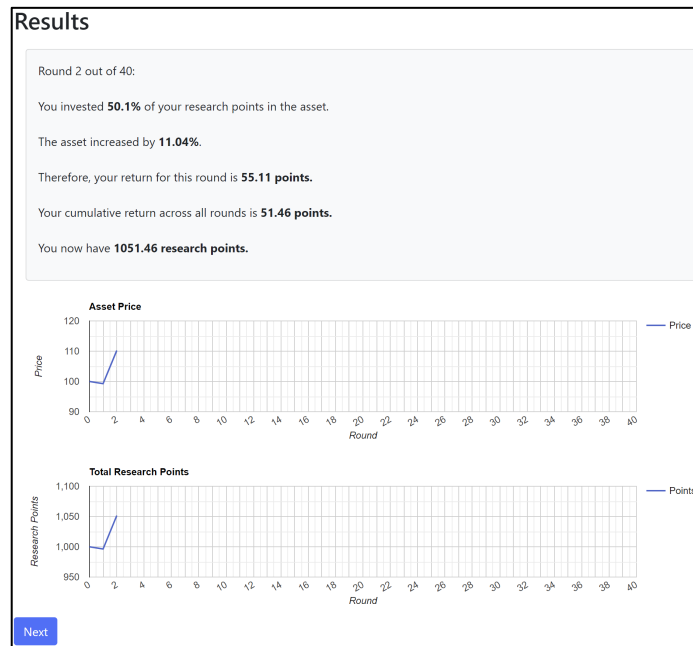


Figure 2: Results Page

After completing all 40 rounds, the total research points (both the initial given endowment and the earned gains or losses) were converted into real dollars and the player was informed of how much they had earned. The experiment is incentivized so that the cumulative return from 10 baseline rounds and 30 principal rounds determine a participant's earnings. We utilize accumulated payoffs rather than allowing each round's return to be independent in order to best mimic a real-world investment context in which a continuous assessment of cumulative gains and losses is standard.

Treatment – Advisor Presentation: The game, as described above, was experimentally manipulated in two ways. The first factor differed the way in which the game’s recommendation system was presented to the player. There were two possible presentations for the system: an algorithmic financial advisor or a human financial advisor. In the algorithm condition, after playing ten practice rounds to learn how the game worked, the player was told:

“These rounds will take the same form as the practice rounds with one exception. Instead of only being able to invest yourself, you can now choose to use the recommendation of a financial investment algorithm.”

Then, on the following page, they are told:

“You will be provided with recommendations, generated by the algorithm, based on the market conditions you are presented with in each round. When the algorithm generates recommendations, it does not know with certainty whether the market is good or bad but invests more when it has high confidence that the market is good and invests less when it has high confidence that the market is bad. Historically, participants who followed its recommendations outperformed those who chose to invest themselves.”

Alternatively, the player was told that a human financial advisor, would be generating recommendations in each round. To control for the quality of the recommendations, the human financial advisor utilized the same backend game system that generated the algorithm’s recommendations and then submitted the given recommendation to the player. This partially mirrors much of the work done by real-life financial advisors in which they utilize a variety of statistical tools and algorithms to make recommendations to their clients. The player was not told

that the human advisor would be using an algorithm to derive their recommendations but the presentation of the advisor was left purposefully vague in order to avoid deceiving the player in any way. The instructions given to the player were identical to those described in the algorithm condition, only changing references of the algorithm to references of the human.

In either condition, at the beginning of each round, players were directed to a waiting page where a short animation of either the human or an algorithm was shown and the player was told that their recommendation was currently being generated (see Figure 3). The inclusion of the waiting page was intended to ensure that timing between rounds remained consistent and player behavior was not impacted by perceptions of the length of time required to make one's own decision relative to using the recommended investment (e.g., players choosing to invest themselves because they wanted to avoid waiting on the advisor). It also improved the realism of the task as the recommendation appeared to take a non-trivial amount of time to be generated.

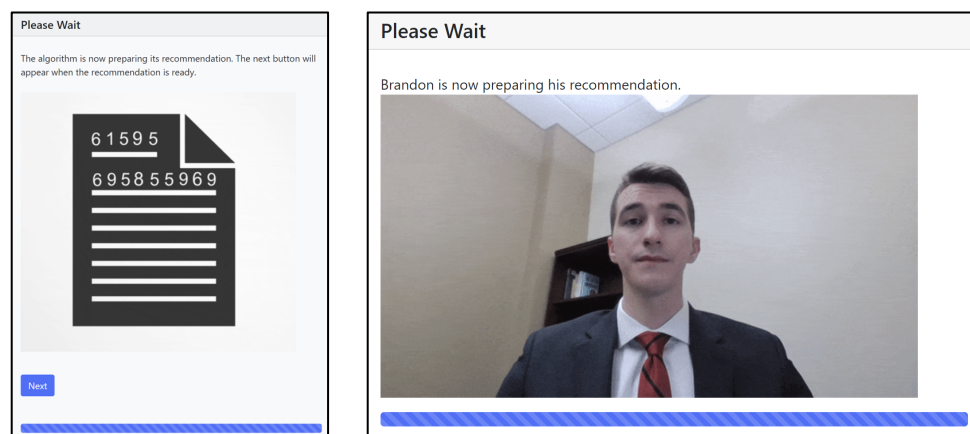


Figure 3: Waiting Page (Left – Algorithm Condition; Right – Human Condition)

Treatment – Disclaimer: The second experimental factor varied across conditions was the inclusion of a disclaimer that came ten rounds after the advisor was introduced. Players were either shown no additional disclaimer, a disclaimer that stated that the advisor's recommendations were occasionally inflated due to an incentive misalignment that encouraged

the advisor to overinvest (betrayal risk), or a disclaimer that again stated that the recommendations were occasionally inflated but instead due to the advisor accidentally overinvesting (error risk).

The intention of the betrayal risk disclaimer was to increase the likelihood that the advisor would violate the trust norms implicit between them and the player. This was meant to elicit betrayal aversion from the player. The player's desire to avoid the emotional cost of being betrayed would then potentially serve as a barrier to advice. However, betrayal aversion is not the sole reason why an individual may not want to take advice in light of an incentive misalignment. If the incentive misalignment artificially inflates investment amounts, the quality of the advice is lowered and players should similarly perceive a decrease in expected return or an increase in their financial risk. In order to isolate only the emotional cost of being betrayed, our error risk disclaimer serves as a pseudo-control in which players should perceive a similar decrease in the quality of the advice without the emotional toll of being betrayed.

Additionally, to further isolate the effects of emotional, non-monetary disutility, both disclaimers again highlighted to the players that historically, the advisor had outperformed participants' own investment decisions, thus ensuring the expected return for the advisor would be higher than the expected return of investing themselves, even in light of the increased risk. A shortened version of the disclaimer was included on the decision page of each round to ensure the saliency of the treatment. The full disclaimers are found in Appendix B. In total, there were six potential variations of the investment game: a human advisor with either no disclaimer (control), a betrayal disclaimer, or an error disclaimer or an algorithmic advisor with the same three disclaimer options.

Finally, to avoid deception, we include logic in the game's recommendation system so that there truly is a small chance of it overinvesting if the player is in one of the two disclaimer conditions. However, in order to control for expected return between conditions, our logic ensures that any overinvesting has an equal likelihood of happening in either a good or bad market state. If it occurs in a good market state, the overinvestment would earn additional revenue for the player. This ensures that the expected return does not change in light of this risk. The same underlying mechanism exists for the error disclaimer condition ensuring the expected return is consistent for every participant.

4.2 Participants & Procedure

Our procedure was intended to mimic the standard procedure one might use in an in-person economics lab adapted to a remote video conferencing platform (e.g., Zoom). The recruitment materials informed potential participants that they would be required to participate in the study through Zoom. Utilizing Zoom to execute the experiment served multiple purposes. First, to avoid using deception in our experiment, we had a real human take part in the game in real-time. Setting up designated time slots with a limited number of participants in each slot allowed us to accomplish this. Second, while mTurk is a valuable platform that allows for a rapid recruitment of diverse participants unmatched by in-person economics labs, issues of data quality and bot accounts limits its value as a research tool. Requiring participants to join via Zoom was a safeguard against bot accounts that gave us higher confidence in the quality of our data.

In total, 460 participants were recruited through Amazon Mechanical Turk (mTurk) to take part in a financial investment game. We anticipated a large number of participants would not attend their session given the requirement to join via Zoom and the safeguards put in place to defend against bot accounts. Of the 460 who signed up, 277 participants subsequently attended

their session and all but two completed the entirety of the procedure. Therefore, 275 participants were included in our final sample. Given that participants were only exposed to their treatment assignments after beginning the game, the conditions of the 183 participants who did not attend their session should be uncorrelated with their decision to be excluded from the sample.

Because the completion of face-to-face tasks over Zoom would not be familiar to mTurk users and may limit their willingness to take part, we offered enticing earnings potential in the experiment. This also increased the stakes of the task for participants, eliciting true decision preferences (opposed to hypothetical or low-stake choices which may or may not align with reality). The game was advertised to make on average \$20-\$30 per hour. The average mTurk requester offers \$11 per hour. After completing our experiment, our average earnings were \$10.29 plus a \$5 show-up fee and the average completion time was 45 minutes. Therefore, our average hourly pay was \$20.34.

Participants first completed a screening survey to sign up for a designated time slot. Utilizing the mTurk API, automated scripts were sent to participants 2 days prior and 1 hour prior to their time slot, the latter of which included the Zoom link that would allow them into their session. Prior to taking place, each session was randomly assigned to either the human or algorithm condition. Conversely, the disclaimer treatment was randomly assigned after participants began playing the game. Each session began with participants joining a Zoom meeting room where they were instructed ahead of time to have changed their display name to their mTurk ID. The experimenter validated that the mTurk IDs listed in the room aligned with those that signed up for the session before providing participants with verbal instructions. In the human condition, the research assistant acting as the financial advisor was introduced using the following statement, “This is Brandon. He is trained in the financial task you are being asked to

complete today and will be aiding in the execution of the experiment.” The research assistant came to each session dressed professionally. Only one individual served in this role for the duration of data collection to ensure no effects of advisor appearance influenced our results. The procedure was the same for the algorithm sessions except that no advisor was introduced.

Participants were then sent a private message through Zoom that contained their personalized link directing them to the browser-based game. Upon clicking the link, they began by going through a brief tutorial that explained the market process and the task that they were being asked to complete. Then they were given the opportunity to invest in 10 unpaid practice rounds where they only had the ability to make their own investment decisions. Following the practice rounds, a short comprehension quiz was completed (see Appendix C). They were then informed that they were being given an endowment of 1000 research points with which to play the game and that this translated to \$5USD.

At the onset of the 40 payoff determining rounds, participants were told of their ability to utilize an advisor as described in the previous section. The human advisor took part in the actual experiment beyond being present in the Zoom session. His interface (see Appendix D) would report the suggested investment, using the same recommendation system as the algorithm, for each participant in each round and would ask him to submit the recommendation to the participant. We kept the design as simple as possible to avoid the possibility of human error while also avoiding deception. Participants may have believed that the advisor was manually developing investment recommendations however they were only ever told that he would be submitting recommendations.

After 10 rounds, they were shown their randomly assigned disclaimer before continuing with the remaining 30 rounds. After completing the game, participants answered a short

demographic survey that asked for their gender, race, ethnicity, education, employment, experience with investing, and familiarity with algorithm design. They also answered exit questions to measure their perceptions of the advisor (see Appendix E). Their final earnings were converted to real dollars and the \$5 show-up fee was added to it before participants were shown their total compensation. Earnings were sent as a bonus payment through mTurk after each session.

4.3 Estimation Approach

Our main dependent variable for our analyses is the participant’s choice to use the advisor’s recommendation or make their own decision in each round. Given the multi-round repeated structure of our experiment and the collinearity between multiple observations from the same experimental participant, we estimate a panel random effects model with robust standard errors and round fixed effects.

$$\text{Use}_{it} = \beta_1 \times \mathbf{Advisor}_i + \beta_2 \times \mathbf{Risk}_i + \beta_3(\mathbf{Advisor}_i \times \mathbf{Risk}_i) + \alpha \times \mathbf{Demographics}_i + \delta_t + \theta_i + u_i \quad (22)$$

The dependent variable, Use_{it} , is a binary indicator that equals 1 if participant i chose to use the advisor in round t and 0 otherwise. $\mathbf{Advisor}_i$ is a vector of binary indicators for the advisor treatment condition assigned to each participant [algorithm vs. human]. $\mathbf{Risk}_i$ is a vector of binary indicators for the risk treatment condition assigned to each participant [no disclaimer vs. error risk vs. betrayal risk]. We also include interactions between advisor presentation and risk treatments. $\mathbf{Demographics}_i$ is a vector of controls capturing heterogeneity in individual demographics (e.g., gender, race, ethnicity, education, work status, experience with algorithms, experience with investing). δ_t includes round fixed effects in order to ensure that any temporal effects are controlled for (e.g., participant learning) and θ_i is the participant-specific random effect. The error term, u_i , is clustered on participant. Estimates on the randomly assigned treatments are assumed unbiased due to the lack of correlation with unobserved individual differences and the error term. This assumption is tested in Section 5.1. This estimation approach allows for a robust

account of time trends in our data and corrects for the nonindependence of multiple observations of the decision to use the advisor from a single participant.

Additionally, we estimate treatment effects for total returns. Since the algorithm, on average, realizes higher returns than participant decisions, any decrease in advisor usage should be accompanied by a decrease in returns as well.

5. Results

Two hundred and seventy-five individuals took part in our financial trading game. Table 1 shows the distribution of participants across treatments in our experiment. We find a roughly equal split for both the additional risk dimension (no disclaimer vs. betrayal risk vs. error risk) and the advisor presentation dimension (algorithm vs. human). The average advisor usage was 56.44% for our baseline rounds and 42.59% in the 30 principal rounds.

	Algorithm	Human	Total
No Disclaimer (Control)	47	42	89
Error Disclaimer	51	42	93
Betrayal Disclaimer	46	47	93
Total	144	131	275

Table 1: Sample Size by Experimental Conditions

Table 2 provides a description of our variables and summary statistics for each. We have a diverse sample of participants with a relatively even split between males and females and racial and ethnic statistics that are comparable to the national averages³. Fifteen participants indicated that they had extensive investment experience and only 7 indicated that they had extensive experience with algorithms.

Variable	Description	Mean (1)	S.D. (2)
Use	The choice to use the advisor in a specific round	0.461	0.498
Earnings	The total earnings from the game	9.141	5.292
Male	Whether the individual is a male	0.455	0.498
Caucasian	Whether the individual is Caucasian	0.749	0.434

³ <https://www.census.gov/quickfacts/fact/table/US/PST045221>

African American	Whether the individual is African American	0.0945	0.293
Asian	Whether the individual is Asian	0.124	0.329
Hispanic	Whether the individual is Hispanic	0.0655	0.247
Some College	Completed some college	0.0982	0.298
Bachelor's Degree	Completed bachelor's degree	0.702	0.457
Advanced Degree	Completed advanced degree	0.200	0.400
Student	Whether the individual is a student	0.0473	0.212
Unemployed	Whether the individual is unemployed	0.135	0.341
Full-Time Employed	Whether the individual is employed full-time	0.0255	0.158
Retired	Whether the individual is retired	0.578	0.494
†Investment Experience	The participant's experience with investing	1.735	0.551
†Algorithm Experience	The participant's experience with algorithms	1.498	0.549

† Likert Scale (1 = No Experience, 3 = Extensive Experience)

Table 2: Summary Statistics

5.1 Balance Checks

Before estimating our main effects, we evaluate the efficacy of random assignment by examining whether there are differences in demographic characteristics across treatment groups. Table 3 provides pair-wise comparisons for all the variables listed in Table 2 across both the advisor presentation factor (Algorithm vs. Human) and the additional risk factor (No Disclaimer vs. Betrayal vs. Error). We conduct 52 pairwise comparisons (13 variables * 4 comparisons) and find that 48 of these comparisons identify insignificant differences between conditions. With an alpha of 0.1, we would expect that 5 comparisons are significant by random chance. We identify only 4 significant differences between these groups. Even so, we control for these variables in our analysis and continue to identify consistent results.

Variable	No Disclaimer (Control) vs. Betrayal	No Disclaimer (Control) vs. Error	Betrayal vs. Error	Algorithm vs. Human
Male	0.663	0.109	0.238	0.180
Caucasian	0.302	0.882	0.233	0.810
African American	0.152	0.901	0.118	0.327
Asian	0.566	0.600	0.268	0.943
Hispanic	0.087*	0.341	0.422	0.780
Some College	0.408	0.609	0.176	0.981
Bachelor's	0.027**	0.140	0.459	0.472
Advanced Degree	0.320	0.243	0.861	0.952
Student	0.132	0.701	0.250	0.913

Full-Time Employed	0.109	0.367	0.011**	0.362
Other Employment	0.322	0.271	0.034**	0.328
Investment Experience	0.989	0.152	0.143	0.298
Algorithm Experience	0.900	0.693	0.583	0.412

Table 3: Comparison of Demographics between Conditions

Next, we leverage our first 10 baseline rounds which precede the introduction of any betrayal or error risk. We evaluate whether any differences exist across our risk conditions in use of the expert advisor and total earnings prior to the introduction of the additional risk. Table 4 estimates the baseline effect of the experimental risk conditions, including participant demographics and round fixed effects (in columns (2) and (4)). Additionally, Table 4 reports the baseline effect of the advisor presentation (human vs. algorithm) for the first 10 rounds.

We do not find any significant differences between risk conditions during the baseline rounds. For example, our primary advisor usage variable sees no significant difference between the betrayal risk treatment and the no disclaimer (control) condition ($p = 0.878$), the betrayal risk treatment and the error risk treatment ($p = 0.866$), or the error risk and the no disclaimer (control) condition ($p = 0.996$). Interestingly, we also see no significant effects on advisor use between the algorithm and human conditions ($p = 0.770$) in these baseline rounds. Those in the algorithm treatment chose to use the advisor, on average, 56.9% of the time while those in the human treatment used the advisor 55.6%. Estimating treatment effects on total earnings for the first 10 rounds similarly provides no significant results. Overall, we find substantial balance across individual characteristics and outcomes prior to the introduction of betrayal.

VARIABLES	(1) Use	(2) Earnings	(3) Use	(4) Earnings
Betrayal Risk	0.00736 (0.0482)	0.998 (1.663)		
Error Risk	-0.000203 (0.0447)	2.722 (1.900)		
Algorithm			0.0110 (0.0376)	-1.017 (1.512)
Constant	0.889***	18.75***	0.888***	20.51***

	(0.127)	(5.146)	(0.125)	(5.191)
Round FE	YES	NO	YES	NO
Demographics	YES	YES	YES	YES
Observations	2,750	275	2,750	275
Number of ID	275	275	275	275

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 4: Baseline Effects – Ten Rounds Prior to Risk Treatment

5.2 Main Effects

Absent the presence of betrayal or error, we find no significant difference in uptake between the human and algorithmic advisor ($p=0.566$). However, this trend changes when the potential for betrayal is introduced. Figure 4 graphs the average advisor usage across each of the treatment sets. First, we find a significant drop in human advisor usage for the betrayal risk treatment (No Disclaimer $\mu=45.1\%$; Error $\mu=45.2\%$; Betrayal $\mu=29.1\%$). However, we see no such drop off in advisor usage for the algorithm treatments (No Disclaimer $\mu=40.6\%$; Error $\mu=48.9\%$; Betrayal $\mu=46.2\%$). The lack of a negative effect in the error conditions implies that the betrayal effect for the human advisor is due to betrayal aversion and not a perceived decrease in expected return or increase in risk generally. Further, this provides preliminary evidence that substituting a human expert with an algorithm may attenuate betrayal aversion⁴.

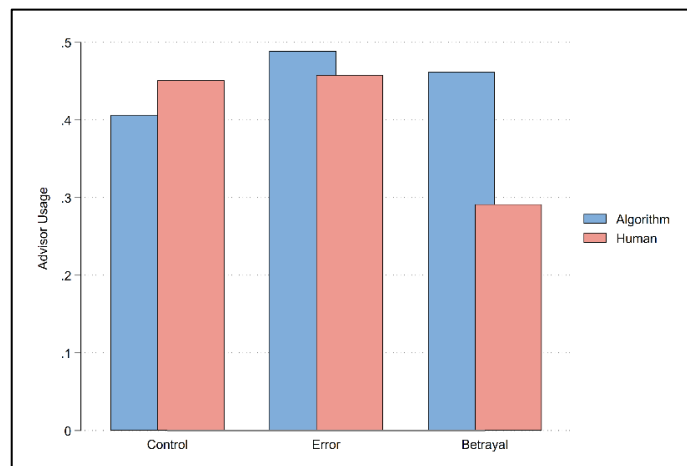


Figure 4: Advisor Usage Across Conditions

⁴ Our visual results show a small increase in advisor utilization for the algorithm treatment when participants are given the error disclaimer. In Table 5, we find that this increase is not statistically significant ($p = 0.451$) and is therefore likely due to random noise.

Our regression model confirms our summary results. The estimates are reported in Table 5. First, we find no significant effect of the algorithm presentation, absent additional betrayal or error risk ($p=0.563$) (column (1)). On average, our estimates show that in the human conditions, adding betrayal risk decreases use by roughly 16% ($p=0.023$) (column (1)). Conversely, introducing risk of an accidental error has no effect on usage ($p=0.924$) (column (1)). This implies that betrayal aversion is observed for the human conditions. Importantly, we find a positive and significant interaction effect of roughly +21% between betrayal risk and the algorithm treatment ($p=0.031$) (column (1)). This implies that the betrayal aversion found in the human treatments is *largely* attenuated when the human advisor is switched out for an algorithm. These effects are consistent when controlling for time fixed effects and participant demographics (columns (2) and (3)).

VARIABLES	(1) Use	(2) Use	(3) Use
Betrayal Risk	-0.159** (0.0702)	-0.159** (0.0704)	-0.155** (0.0706)
Error Risk	0.00714 (0.0753)	0.00714 (0.0754)	0.00841 (0.0756)
Algorithm	-0.0444 (0.0768)	-0.0444 (0.0770)	-0.0284 (0.0781)
Algorithm $\times$ Betrayal	0.215** (0.0992)	0.215** (0.0994)	0.178* (0.101)
Algorithm $\times$ Error	0.0754 (0.103)	0.0754 (0.103)	0.0680 (0.101)
Constant	0.451*** (0.0563)	0.421*** (0.0602)	0.600*** (0.146)
Round FE	NO	YES	YES
Demographics	NO	NO	YES
Observations	8,250	8,250	8,250
Number of ID	275	275	275

Robust standard errors in parentheses

*** $p<0.01$, ** $p<0.05$, * $p<0.1$

Table 5: Main Effect of Treatment on Use

5.2.1 Empirical Extensions

To further enrich our understanding of the impact of algorithmic experts on betrayal aversion, we explore a number of empirical extensions.

Persistence of Treatment Effects: Much of the betrayal aversion literature has utilized one-shot trust games to measure its effect. To our knowledge, no work has utilized a repeated game in this way and therefore what is known of the persistence of betrayal aversion over time is limited. One potential explanation is that participants do experience betrayal aversion towards the algorithm but that aversion dissipates quickly, leading to a null effect. Therefore, the following analyses explore heterogeneity across time for our treatment effects. Figure 5a and 5b plot average advisor usage across rounds for each condition. In Figure 5a, we can clearly see the effect of betrayal for the human conditions. It appears as if the effect remains relatively constant across rounds. In Figure 5b, algorithm usage rates for the betrayal treatment appear to drop initially before rising by the tenth principal round. This could imply that the algorithm experienced some initial betrayal effects but, unlike the human conditions, those effects dissipated quickly.

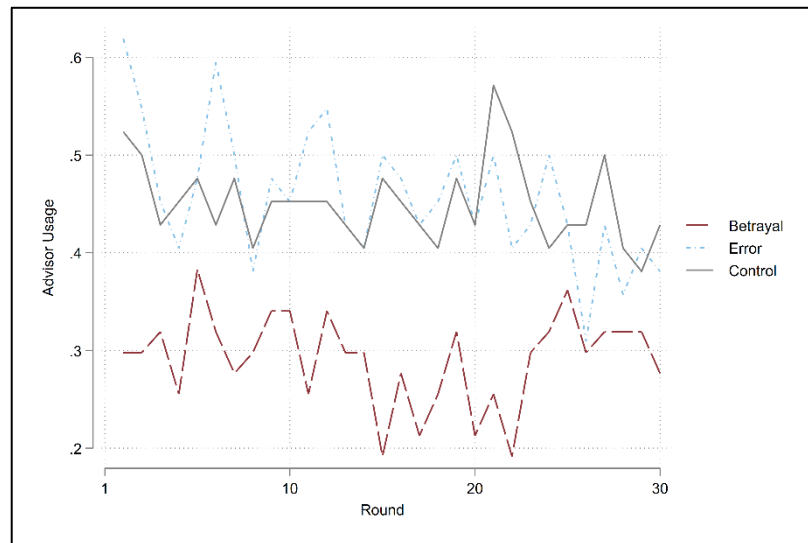


Figure 5a: Human Usage Rates Across Treatments

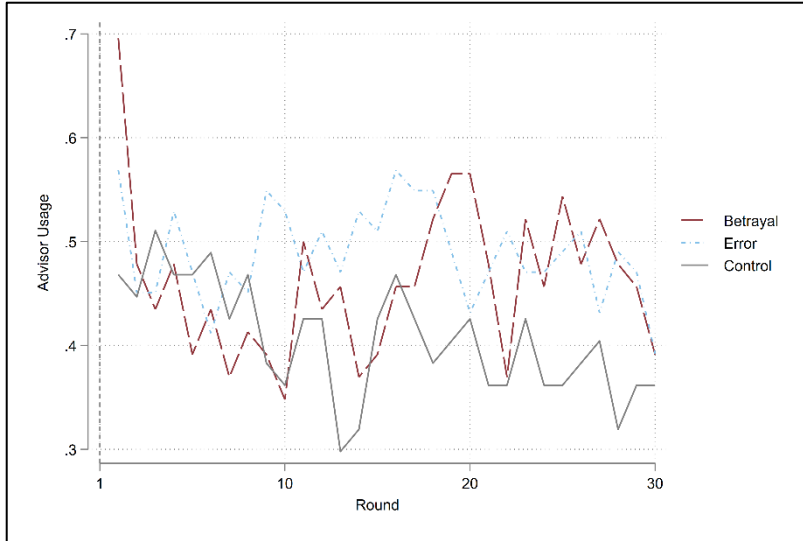


Figure 5b: Algorithm Usage Rates Across Treatments

Table 5 above confirms these results. We estimate our model for the first 10 principal rounds (1-10) (column (4)), the next 10 principal rounds (11-20) (column (5)), and the last 10 principal rounds (21-30) (column (6)). While betrayal aversion towards the human advisor remains consistent across rounds, we find that in the first 10 treated rounds, substituting the human with an algorithm does not fully attenuate the betrayal aversion ($p=0.266$) (column (3)). However, after at least 10 treated rounds, the betrayal aversion towards algorithms is completely attenuated ($p=0.058$) (column (4)). As stated above, this provides evidence of some initial, but not persistent, betrayal aversion for algorithms.

	1-10 (4) Use	11-20 (5) Use	21-30 (6) Use
VARIABLES			
Betrayal Risk	-0.147** (0.0708)	-0.163** (0.0772)	-0.161* (0.0823)
Error Risk	0.0310 (0.0745)	0.0466 (0.0864)	-0.0492 (0.0866)
Algorithm	-0.0106 (0.0770)	-0.0156 (0.0859)	-0.0651 (0.0885)
Algorithm $\times$ Betrayal	0.141 (0.101)	0.206* (0.110)	0.214* (0.116)
Algorithm $\times$ Error	0.00835 (0.104)	0.0544 (0.115)	0.140 (0.117)
Constant	0.547*** (0.0598)	0.618*** (0.167)	0.709*** (0.166)
Observations	2,750	2,750	2,750

Number of ID	275	275	275
Robust standard errors in parentheses			
*** p<0.01, ** p<0.05, * p<0.1			

Table 6: Persistence of Treatment Effects

Prior Advisor Performance: While advisor performance is assumed to be randomly distributed across treatment conditions, we examine the impact of prior performance on advisor utilization to better understand the relationship that betrayal aversion has with monetary considerations. Table 7 estimates the same estimation model specified in (22) with the addition of the lagged investment return interacted with the participant's advisor utilization decision in the prior round.

The estimate on the lagged return is negative and statistically significant ($p = 0.000$), implying that if the participant invested without help in the prior round and performed well, they were less likely to use the advisor in the current round. Likewise, the estimate on the interaction term is positive and statistically significant ($p = 0.000$) implying that if the advisor performed well for the participant in the prior round, they were more likely to utilize the advisor in the current round. Interestingly, while, as one would expect, financial returns do have an effect on advisor utilization, the effect of betrayal aversion remains consistent. This implies that the prior performance of the advisor is orthogonal to our central research question and provides further support for our claim that the effect of betrayal aversion we measure is not a response to monetary considerations or concerns.

VARIABLES	(1) Use
Betrayal Risk	-0.157** (0.0690)
Error Risk	0.00810 (0.0741)
Algorithm	-0.0443 (0.0753)
Algorithm $\times$ Betrayal	0.212** (0.0975)
Algorithm $\times$ Error	0.0713 (0.101)
Prior Round Return	-0.000279*** (0.00005)
Prior Round Return $\times$	0.000645***

Prior Round Advisor Use	(0.00008)
Constant	0.463*** (0.0580)
Observations	8,250
Number of ID	275

Robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 7: Prior Advisor Performance

Further, we estimate the model with the inclusion of all interaction terms to determine if our betrayal treatment amplifies or diminishes any effects of prior advisor performance. We still find a statistically significant negative effect of betrayal aversion, although lower in degree, for the human condition ($\beta = -0.0673, p = 0.032$). Similarly, our estimates again confirm that replacing the human advisor with an algorithm attenuates this betrayal aversion ($\beta = 0.0944, p = 0.034$). Interestingly, we find no statistically significant interactions between our betrayal risk treatment and prior advisor performance. This again provides further compelling evidence that the effects we measure are non-monetary in nature.

Finally, we estimate the model with an indicator that reflects whether the participant's first use of the advisor was positive or negative to determine if a path dependency exists that may differ across treatments. Our results report no significant interactions between our first exposure indicator and our treatments. The estimates for both of the preceding two models can be found in Appendix F and Appendix G respectively.

Financial Consequences of Betrayal Aversion: We estimate betrayal treatment effects for the total earnings received by each participant (Table 6). Given that the human advisor uses the algorithm to make decisions and there is no difference in expected return for the betrayal or error conditions, if a participant chose the advisor over investing themselves in any condition, we would expect the same average returns. However, given the lack of advisor usage in the human betrayal condition, we analyze the effect of this decreased uptake on subsequent earnings. Our results indicate a \$2.15 decrease in total earnings for those

in betrayal treatment with a human advisor ($p=0.072$). We observe no significant effects of the error conditions for either advisor (Human $p=0.195$; Algorithm $p=0.281$) nor is there an effect of betrayal risk for the algorithm treatment ($p=0.883$). This supports the conclusion that the decrease in earnings is due to participants choosing to utilize the advisor less in the human betrayal condition. Given that the average earnings for those in the no disclaimer condition with a human advisor was \$10.29 (not including the \$5 show up fee), a decrease of \$2.15 would equate to a 21% drop in total earnings. These findings highlight the real financial consequences of betrayal aversion.

VARIABLES	Human (1) Earnings	Algorithm (2) Earnings
Betrayal Risk	-2.153* (1.186)	0.180 (1.223)
Error Risk	-1.502 (1.153)	-1.118 (1.032)
Constant	10.29*** (0.933)	9.577*** (0.835)
Observations	131	144
R-squared	0.030	0.012

Robust standard errors in parentheses

*** $p<0.01$, ** $p<0.05$, * $p<0.1$

Table 8: Betrayal Effect on Total Earnings

Exit Questions: In Table 9 we estimate the effect of our treatments on a number of exit questions related to feelings of being misled, having trust violated, and feeling betrayed. Each of these questions are meant to reflect variations in our definition of betrayal (a violation of a trust norm) and therefore aide in confirming our phenomenon of interest. The questions were answered with a 5-point Likert scale where 1 = strongly disagree and 5 = strongly agree. Positive coefficients imply a stronger trend towards agreeing with the question. Column 1 reports estimates for responses to the question, “If you chose to invest with the advisor, you were concerned about being misled.” For the human condition, betrayal has a positive and statistically significant effect ($p=0.006$). This effect is almost entirely cancelled out by the algorithm

and betrayal interaction term ($p=0.048$) implying that feelings of being misled from the betrayal risk were not a concern in the algorithm conditions. Column 2 reports estimates to the question asking if participants were concerned about their trust being violated. Again, we find similar results to the first question with the caveat that the algorithm interaction term is slightly insignificant ($p=0.103$).

VARIABLES	(1) Misled	(2) Trust Violated	(3) Feel Betrayed
Algorithm	0.245 (0.242)	0.0755 (0.236)	0.390 (0.250)
Betrayal Risk	0.671*** (0.240)	0.650** (0.252)	0.815*** (0.254)
Error Risk	0.0476 (0.256)	-0.0476 (0.249)	0.214 (0.264)
Algorithm $\times$ Betrayal	-0.663** (0.333)	-0.568 (0.347)	-1.095*** (0.357)
Algorithm $\times$ Error	0.0825 (0.339)	0.164 (0.338)	-0.185 (0.358)
Constant	3.095*** (0.183)	2.690*** (0.175)	2.738*** (0.177)
Observations	275	275	275
R-squared	0.038	0.038	0.046

Robust standard errors in parentheses

*** $p<0.01$, ** $p<0.05$, * $p<0.1$

Table 9: Exit Question Analysis

Finally, participants were asked if they had concerns about feeling betrayed by the advisor. The effect of the treatment in Column 3 is the largest coefficient ($p=0.001$). This implies that our treatment worked as intended and elicited feelings of betrayal in participants. Additionally, the interaction term for the algorithm condition cancels out the treatment effect entirely ($p=0.002$), again providing evidence that algorithms can attenuate betrayal aversion.

6. Robustness

6.1 Experiment 2 – Robustness to Sample & Advisor

To validate the results of our first experiment, we conducted a condensed version of the investment game with a new sample: undergraduate students. Unlike the main study, we do not include an

additional error treatment (only a betrayal risk treatment and the no disclaimer condition). Finally, there is a within-subjects fee treatment that is introduced after the first 20 rounds which decreased advisor use to almost zero across conditions. Therefore, we chose to analyze only the first 20 rounds where there were no fees. All other procedures are identical to the main study.

There are several key benefits to this follow-up study. First, we utilize an undergraduate economics lab to recruit our sample. Researchers that recruit through this lab are prohibited from using any form of deception in their experiments. This is explicitly highlighted to participants, and they should therefore understand that any claims made throughout the experiment are true. This additional confidence in the veracity of information provided by the researchers help alleviate concerns around the believability of the information provided to participants in the first experiment. Additionally, we utilized a different human financial advisor to further provide robustness around advisor appearance and helps alleviate concerns of effects specific to a particular individual.

One hundred and twenty-three participants took part in our experiment. Table 10 shows the results of our panel random effects models that we used to estimate our treatment effects. Our effects from the main study are replicated here. Column 1 estimates a roughly 12% decrease in advisor usage when betrayal risk is introduced ($p = 0.091$). Conversely, we see no significant decrease in advisor usage when betrayal risk is introduced to the algorithm conditions ($p = 0.167$). Column 3 reports the interaction term between betrayal risk and the algorithm, confirming that switching the advisor from a human to an algorithm counteracts the measured betrayal aversion ($p = 0.026$).

Likewise, Column 4 shows that betrayal aversion decreased total earnings by roughly \$2.13 ($p = 0.003$), strikingly similar to the decrease found in the main study. Those in the algorithm conditions saw no such decrease in earnings ($p = 0.387$, column (5)). The results from Experiment 2 confirm findings from Experiment 1 across the board. This level of robustness is difficult to attain even when replicating experimental procedures that draw from the same population. Provided that Experiment 2 drew from an entirely new population with large demographic differences (e.g., college-aged students versus a far older

population), this replication adds confidence to the robustness of our original findings.

VARIABLES	Human (1) Use	Algorithm (2) Use	All (3) Use	Human (4) Earnings	Algorithm (5) Earnings
Betrayal Risk	-0.123* (0.0712)	0.0774 (0.0561)	-0.123* (0.0708)		
Algorithm			-0.118* (0.0697)	-2.128*** (0.709)	-0.566 (0.651)
Algorithm $\times$ Betrayal			0.201** (0.0902)		
Constant	0.423*** (0.0560)	0.305*** (0.0421)	0.423*** (0.0557)	18.72*** (0.593)	17.62*** (0.555)
Number of ID	48	75	123	48	75

Robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 10: Experiment 2 Results

6.2 Experiment 3 – Robustness to Context

To explore the presence of betrayal aversion, and the ability of algorithms to attenuate it, beyond the financial trading context, we conducted a follow-up study that presented participants with a series of ten scenarios in which a violation of trust norms was apparent (a betrayal). Each scenario presented this phenomenon in a different context. These included financial investment, product recommendations, customer support, recruitment and hiring, education, automotive manufacturing, healthcare, insurance, social media content moderation, and delivery operations. The full written scenarios can be found in Appendix H.

For each scenario, participants were randomly assigned to read a version of the scenario in which the actor was either a human or an algorithm. We then ask a series of questions meant to gauge 1) the participant's belief on the amount of objective harm inflicted on the impacted party and 2) potential feelings of betrayal. The full list of questions can be found in Appendix I.

The results are shown in the table below and in Table 11. We regress the algorithm treatment, context fixed effects, and the interaction between context and the algorithm treatment on the assigned amount of damages (a measure of objective harm), the level of blame assigned to the actor (a measure of

intentionality), and the degree of feeling betrayed (our phenomenon of interest). Column 1 reports the assigned amount of damages. These results highlight that perceptions of the objective harm inflicted on the impacted party are consistent across contexts irrespective of the actor being a human or an algorithm. This is equivalent to our claim in the first experiment that the perceptions of monetary objective returns are controlled for.

Column 2 reports differences in blame attribution. The coefficient for our algorithm treatment is indicative of the difference in blame attribution for the first context (financial services). There is a statistically significant negative effect of the treatment, implying that when shown the scenario in which the actor was an algorithm, the amount of attributed blame was significantly lower. In nine out of ten contexts, the interaction terms are insignificant, implying that the difference in blame attribution between the human and algorithmic actor are consistent with what was found for the first context. Social media content moderation is the only context with a statistically significant interaction term. However, this effect is similarly negative, implying that the difference in blame attribution was even wider for this context. These results highlight the lack of intentionality, a common precursor to betrayal aversion, attributed to algorithms.

Finally, Column 3 reports differences in reported feelings of betrayal. For the first context, there is a negative and statistically significant effect of the algorithm treatment, implying lower feelings of betrayal in the algorithm treatment. This effect remains consistent for all but three contexts (customer support, education, and insurance). These three contexts have a significant interaction term that cancels out the initial attenuation effect. One potential reason for this could be that the harm caused in these contexts was severe enough for the participants to see beyond the algorithm and recognize the betrayal coming from those responsible for the algorithm's development and use.

These results provide preliminary evidence that our findings from Experiment 1 may persist beyond financial services. We find that participants in a variety of contexts feel less betrayed by an algorithm than they do a human who imparts the same degree of harm, despite the fact that the algorithms are designed and deployed by humans.

VARIABLES	(1) Damage	(2) Blame	(3) Betrayal
Algorithm	15.72 (34.61)	-8.986** (3.647)	-0.515*** (0.176)
Alg × Retail	33.03 (53.59)	5.320 (5.453)	0.471 (0.362)
Alg × Cust Support	62.12 (48.53)	5.017 (4.172)	0.589** (0.278)
Alg × Recruitment	-27.02 (49.52)	-3.625 (5.139)	0.209 (0.287)
Alg × Education	-45.74 (47.88)	-2.039 (4.852)	0.513** (0.253)
Alg × Automotive	-55.35 (46.33)	1.163 (4.193)	0.314 (0.260)
Alg × Healthcare	-67.66 (45.86)	-0.0312 (4.376)	0.600* (0.315)
Alg × Insurance	41.36 (45.47)	7.142 (4.875)	0.758*** (0.256)
Alg × Content Mod	-16.44 (58.09)	-9.785** (4.812)	0.0953 (0.268)
Alg × Delivery	0.768 (48.23)	1.246 (4.505)	0.418 (0.268)
Constant	628.7*** (25.04)	81.79*** (2.736)	6.095*** (0.124)
Context FE	Yes	Yes	Yes
Observations	1247	1,250	1,248
Number of IDs	125	125	125

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 11: Experiment 3 Results

7. Discussion: Implications for Research and Practice

Our results show that while human experts elicit significant betrayal aversion, substituting those experts with algorithms may attenuate most, if not all, of the effect. Additionally, we highlight the real financial consequences of betrayal aversion and show a 20% decrease in earnings due to betrayal. Finally, we show that while algorithms may experience some initial betrayal aversion, it quickly subsides, whereas the aversion persists for human experts.

This work is not without limitations. First, our main study only focus on the context of financial investing. Future work could explore this avenue by studying contexts with higher stakes and exploring the effect that stakes have on betrayal aversion and algorithm aversion. Our third experiment opens the door to further exploring betrayal aversion across contexts, providing preliminary evidence for its presence in a variety of domains. Additionally, the experiment is not able to measure individual level betrayal due to the between-subjects design. Aimone et al. (2015) propose a betrayal aversion elicitation task that uses a within-subjects design to determine betrayal aversion at the individual level. Future research could adopt this framework to uncover heterogeneity around individual characteristics and their impact on betrayal aversion. Further, this paper only considers the role of betrayal aversion as it relates to algorithms. In reality, the choice to use an algorithm is likely influenced by a number of other factors including familiarity with the algorithm provider, transparency of the algorithm, autonomy over the algorithm, and others. Future research could incorporate these ideas and explore their initial effects and interactive effects with betrayal aversion on algorithm adoption and use.

Despite these limitations, our work has significant implications for research. Our findings highlight the value in looking to behavioral and experimental economics to uncover potential attributes of algorithmic tools that drive an individual to utilize an expert advisor. Further, we show that studying the adoption of algorithms and the interactions that humans have with them should not be viewed as an isolated phenomenon. Instead, by comparing individual behavior with other humans versus an algorithm, we can better understand what mechanisms drive adoption. For example, one possible explanation for the attenuation of betrayal aversion by algorithms is that participants in the human condition were able to interact with their advisor. This element of social connection heightens trust, which subsequently increases concerns over betrayal. Researchers can use these findings to develop solutions to hesitation around accepting algorithmic tools that utilize the inherent strengths of the algorithm itself. Further, future research could extend this potential pathway by varying the levels of social connection that an individual has with an algorithm.

Finally, our findings highlight the need for behavioral economists to revisit previously well-accepted phenomena in light of advancing technologies. While the majority view of the literature is that inanimate objects can elicit betrayal aversion, we provide evidence to the contrary. One potential reason for this deviation could be that past work considering inanimate objects and betrayal aversion focused exclusively on products meant to protect consumer's *physical* safety. It may be the case that while algorithms can attenuate betrayal aversion in a financial services setting, they may still elicit the fear when a consumer's physical outcomes are at risk. Future work can extend this thought by looking for instances of betrayal aversion with smart cities and Internet-of-Things devices that direct traffic or deploy emergency services, betrayal aversion elicited through online health portals, or the effect of algorithmic vaccine deployment on betrayal fears. Each of these pathways would prove fruitful in developing a deeper understanding of human decision-making and thought.

Industry could also benefit from the findings we present. Largely, firms across industries are adopting algorithmic solutions and determining which roles within their corporations could be augmented by algorithms. While efficiencies and improved operations play a large role in this consideration, acceptance from shareholders and customers is equally important. Our work provides valuable insight into the relationship humans have with algorithms and how that relationship leads to adoption and acceptance. For example, firms that provide face-to-face services that rely on consumer acceptance of their output may benefit from enhanced algorithmic participation, given the decrease in social expectations of algorithms from the consumer.

More specifically, financial services can utilize our results to improve uptake of financial planning services. While firms are increasingly introducing algorithmic tools to industry customers, there is little work showing the value of these tools to individual consumers. By augmenting traditional financier involvement in strategic customer planning with advanced algorithmic tools, consumers may be more willing to seek expert advice.

Overall, our results highlight the potential of algorithms to decrease the trust burden that many consumers face with “experts” that they encounter. Mitigating betrayal aversion can lead to increased demand for expert advice and better individual outcomes for consumers. The potential of algorithms from the perspective of efficiency and accuracy is only one piece of the larger puzzle. Algorithms may make individuals feel at ease and allow them to overlook some of the less impactful social rules that might have restricted their previous gains.

8. References

- Aimone, J., Ball, S., and King-Casas, B. 2015. "The Betrayal Aversion Elicitation Task: An Individual Level Betrayal Aversion Measure," *PLOS ONE* (10:9), p. e0137491.
- Aimone, J. A., and Houser, D. 2012. "What You Don't Know Won't Hurt You: A Laboratory Analysis of Betrayal Aversion," *Experimental Economics* (15:4), pp. 571-588.
- Aimone, J. A., Houser, D., and Weber, B. 2014. "Neural Signatures of Betrayal Aversion: An Fmri Study of Trust," *Proceedings of the Royal Society B: Biological Sciences* (281:1782), p. 20132127.
- Al-Natour, S., and Benbasat, I. 2009. "The Adoption and Use of It Artifacts: A New Interaction-Centric Model for the Study of User-Artifact Relationships," *J. AIS* (10).
- Alexander, V., Blinder, C., and Zak, P. J. 2018. "Why Trust an Algorithm? Performance, Cognition, and Neurophysiology," *Computers in Human Behavior* (89), pp. 279-288.
- Alt, R., Beck, R., and Smits, M. T. 2018. "Fintech and the Transformation of the Financial Industry." Springer, pp. 235-243.
- Arrow, K. J. 1971. "The Theory of Risk Aversion," *Essays in the theory of risk-bearing*, pp. 90-120.
- Beck, A. H., Sangoi, A. R., Leung, S., Marinelli, R. J., Nielsen, T. O., van de Vijver, M. J., West, R. B., van de Rijn, M., and Koller, D. 2011. "Systematic Analysis of Breast Cancer Morphology Uncovers Stromal Features Associated with Survival," *Sci Transl Med* (3:108), p. 108ra113.
- Bell, D., and Gana, L. 2012. "Algorithmic Trading Systems: A Multifaceted View of Adoption," *2012 45th Hawaii International Conference on System Sciences: IEEE*, pp. 3090-3099.
- Benbasat, I., and Wang, W. 2005. "Trust in and Adoption of Online Recommendation Agents," *J. AIS* (6).
- Berger, B., Adam, M., Rühr, A., and Benlian, A. 2021. "Watch Me Improve—Algorithm Aversion and Demonstrating the Ability to Learn," *Business & Information Systems Engineering* (63:1), pp. 55-68.
- Bernard, C., Chen, J. S., and Vanduffel, S. 2015. "Rationalizing Investors' Choices," *Journal of Mathematical Economics* (59), pp. 10-23.

- Birnberg, J., and Zhang, Y. 2010. "When Betrayal Aversion Meets Loss Aversion: The Effects of Changes in Economic Conditions on Internal Control System Choices," *Journal of Management Accounting Research* (23).
- Bohnet, I., Greig, F., Herrmann, B., and Zeckhauser, R. 2008. "Betrayal Aversion: Evidence from Brazil, China, Oman, Switzerland, Turkey, and the United States," *American Economic Review* (98:1), pp. 294-310.
- Bohnet, I., Herrmann, B., and Zeckhauser, R. 2010. "Trust and the Reference Points for Trustworthiness in Gulf and Western Countries," *The Quarterly Journal of Economics* (125:2), pp. 811-828.
- Bohnet, I., and Zeckhauser, R. 2004. "Trust, Risk and Betrayal," *Journal of Economic Behavior & Organization* (55:4), pp. 467-484.
- Brogaard, J., Hendershott, T., and Riordan, R. 2014. "High-Frequency Trading and Price Discovery," *The Review of Financial Studies* (27:8), pp. 2267-2306.
- Burton, J. W., Stein, M.-K., and Jensen, T. B. 2020. "A Systematic Review of Algorithm Aversion in Augmented Decision Making," *Journal of Behavioral Decision Making* (33:2), pp. 220-239.
- Cao, L., Yuan, G., Leung, T., and Zhang, W. 2020. "Special Issue on Ai and Fintech: The Challenge Ahead," *IEEE Intelligent Systems* (35:2), pp. 3-6.
- Castelo, N., Bos, M. W., and Lehmann, D. R. 2019. "Task-Dependent Algorithm Aversion," *Journal of Marketing Research* (56:5), pp. 809-825.
- Chakrabarty, B., Moulton, P., and Wang, X. 2015. "Attention Effects in a High-Frequency World,").
- Chen, D. L., Schonger, M., and Wickens, C. 2016. "Otree—an Open-Source Platform for Laboratory, Online, and Field Experiments," *Journal of Behavioral and Experimental Finance* (9), pp. 88-97.
- Colarelli, S. M., and Thompson, M. 2008. "Stubborn Reliance on Human Nature in Employee Selection: Statistical Decision Aids Are Evolutionarily Novel," *Industrial and Organizational Psychology* (1:3), pp. 347-351.
- Dawes, R. M. 1979. "The Robust Beauty of Improper Linear Models in Decision Making," *American Psychologist* (34:7), pp. 571-582.
- Diab, D., Pui, S.-Y., Yankelevich, M., and Highhouse, S. 2011. "Lay Perceptions of Selection Decision Aids in Us and Non-Us Samples," *International Journal of Selection and Assessment* (19).
- Dietvorst, B. J., Simmons, J. P., and Massey, C. 2015. "Algorithm Aversion: People Erroneously Avoid Algorithms after Seeing Them Err," *Journal of Experimental Psychology: General* (144:1), pp. 114-126.
- Dietvorst, B. J., Simmons, J. P., and Massey, C. 2018. "Overcoming Algorithm Aversion: People Will Use Imperfect Algorithms If They Can (Even Slightly) Modify Them," *Management Science* (64:3), pp. 1155-1170.
- Eastwood, J., Snook, B., and Luther, K. 2012. "What People Want from Their Professionals: Attitudes toward Decision-Making Strategies," *Journal of Behavioral Decision Making* (25).
- Filiz, I., Judek, J. R., Lorenz, M., and Spiwoks, M. 2021. "Reducing Algorithm Aversion through Experience," *Journal of Behavioral and Experimental Finance*), p. 100524.

- Ge, R., Zheng, Z., Tian, X., and Liao, L. 2021. "Human–Robot Interaction: When Investors Adjust the Usage of Robo-Advisors in Peer-to-Peer Lending," *Information Systems Research*).
- Gomber, P., Kauffman, R. J., Parker, C., and Weber, B. W. 2018. "Financial Information Systems and the Fintech Revolution." Taylor & Francis, pp. 12-18.
- Grégoire, Y., and Fisher, R. J. 2008. "Customer Betrayal and Retaliation: When Your Best Customers Become Your Worst Enemies," *Journal of the Academy of Marketing Science* (36:2), pp. 247-261.
- Hendershott, T., Jones, C. M., and Menkveld, A. J. 2011. "Does Algorithmic Trading Improve Liquidity?," *The Journal of Finance* (66:1), pp. 1-33.
- Hendershott, T., Zhang, X., Zhao, J. L., and Zheng, Z. 2021. "Fintech as a Game Changer: Overview of Research Frontiers," *Information Systems Research* (32:1), pp. 1-17.
- Highhouse, S. 2008. "Stubborn Reliance on Intuition and Subjectivity in Employee Selection," *Industrial and Organizational Psychology: Perspectives on Science and Practice* (1:3), pp. 333-342.
- Holt, C. A., and Laury, S. K. 2002. "Risk Aversion and Incentive Effects," *The American Economic Review* (92:5), pp. 1644-1655.
- Hong, K., and Bohnet, I. 2007. "Status and Distrust: The Relevance of Inequality and Betrayal Aversion," *Journal of Economic Psychology* (28:2), pp. 197-213.
- Jussupow, E., Spohrer, K., Heinzl, A., and Gawlitza, J. 2021. "Augmenting Medical Diagnosis Decisions? An Investigation into Physicians' Decision-Making Process with Artificial Intelligence," *Information Systems Research*).
- Koehler, J. J., and Gershoff, A. D. 2003. "Betrayal Aversion: When Agents of Protection Become Agents of Harm," *Organizational Behavior and Human Decision Processes* (90:2), pp. 244-261.
- Kou, G. 2019. "Introduction to the Special Issue on Fintech." SpringerOpen, pp. 1-3.
- Logg, J. M., Minson, J. A., and Moore, D. A. 2019. "Algorithm Appreciation: People Prefer Algorithmic to Human Judgment," *Organizational Behavior and Human Decision Processes* (151), pp. 90-103.
- Longoni, C., Bonezzi, A., and Morewedge, C. K. 2019. "Resistance to Medical Artificial Intelligence," *Journal of Consumer Research* (46:4), pp. 629-650.
- Lourenço, C. J. S., Dellaert, B. G. C., and Donkers, B. 2020. "Whose Algorithm Says So: The Relationships between Type of Firm, Perceptions of Trust and Expertise, and the Acceptance of Financial Robo-Advice," *Journal of Interactive Marketing* (49), pp. 107-124.
- Nass, C., and Moon, Y. 2000. "Machines and Mindlessness: Social Responses to Computers," *Journal of Social Issues* (56:1), pp. 81-103.
- Nass, C., Steuer, J., and Tauber, E. R. 1994. "Computers Are Social Actors," *Proceedings of the SIGCHI conference on Human factors in computing systems*, pp. 72-78.
- Önkal, D., Goodwin, P., Thomson, M., Gönül, S., and Pollock, A. 2009. "The Relative Influence of Advice from Human Experts and Statistical Methods on Forecast Adjustments," *Journal of Behavioral Decision Making* (22:4), pp. 390-409.
- Orlikowski, W. J., and Scott, S. V. 2015. "The Algorithm and the Crowd: Considering the Materiality of Service Innovation," *MIS Q.* (39), pp. 201-216.
- Parker, G., Van Alstyne, M., and Jiang, X. 2017. "Platform Ecosystems:: How Deve/Operators Invert the Firm," *MIS Quarterly* (41:1), pp. 255-266.

- Prahl, A., and Van Swol, L. 2017. "Understanding Algorithm Aversion: When Is Advice from Automation Discounted?," *Journal of Forecasting* (36:6), pp. 691-702.
- Pratt, J. W. 1964. "Risk Aversion in the Small and in the Large," *Econometrica* (32:1/2), pp. 122-136.
- Qiu, L., and Benbasat, I. 2005. "Online Consumer Trust and Live Help Interfaces: The Effects of Text-to-Speech Voice and Three-Dimensional Avatars," *International Journal of Human-Computer Interaction* (19:1), pp. 75-94.
- Ransbotham, S., Gerbert, P., Reeves, M., Kiron, D., and Spira, M. 2018. "Artificial Intelligence in Business Gets Real," MIT Sloan Management Review and The Boston Consulting Group.
- Reeves, B., and Nass, C. I. 1996. *The Media Equation: How People Treat Computers, Television, and New Media Like Real People and Places*. New York, NY, US: Cambridge University Press.
- Schanke, S., Burtch, G., and Ray, G. 2021. "Estimating the Impact of “Humanizing” Customer Service Chatbots," *Information Systems Research* (32:3), pp. 736-751.
- Scherer, L. D., de Vries, M., Zikmund-Fisher, B. J., Witteman, H. O., and Fagerlin, A. 2015. "Trust in Deliberation: The Consequences of Deliberative Decision Strategies for Medical Decisions," *Health Psychol* (34:11), pp. 1090-1099.
- Seiders, K., Flynn, A. G., Berry, L. L., and Haws, K. L. 2015. "Motivating Customers to Adhere to Expert Advice in Professional Services: A Medical Service Context," *Journal of Service Research* (18:1), pp. 39-58.
- Shnoor, B. 2009. "Loss of Chance: A Behavioral Analysis of the Difference between Medical Negligence and Toxic Torts,").
- Sparrow, B., Liu, J., and Wegner Daniel, M. 2011. "Google Effects on Memory: Cognitive Consequences of Having Information at Our Fingertips," *Science* (333:6043), pp. 776-778.
- Tetlock, P. E., and Gardner, D. 2016. *Superforecasting: The Art and Science of Prediction*. Random House.
- Van Swol, L. M. 2011. "Forecasting Another’s Enjoyment Versus Giving the Right Answer: Trust, Shared Values, Task Effects, and Confidence in Improving the Acceptance of Advice," *International Journal of Forecasting* (27:1), pp. 103-120.
- Venkatesh, V., and Davis, F. 2000. "A Theoretical Extension of the Technology Acceptance Model: Four Longitudinal Field Studies," *Management Science* (46), pp. 186-204.
- Wilson, H., Daugherty, P., and Bianzino, N. 2017. "When Ai Becomes the New Face of Your Brand," *Harvard Business Review* (27).
- Woodhouse, E. J., and Nieusma, D. 1997. "When Expert Advice Works, and When It Does Not," *IEEE Technology and Society Magazine* (16:1), pp. 23-29.
- Zhang, X. 2020. "Experiential and Neurobiological Influences on Economic Preferences and Risky Decision Making." Virginia Tech.

9. Appendices

Appendix A: Market & Algorithm Design

The risky asset in our experimental market follows a two-state (good or bad) Markov-switching Gaussian random walk with a state switching probability of 35%⁵, adapted from Zhang (2020)). In other words, if the previous round was a good state, there is a 35% chance that the following round would be a bad state and a 65% that it would be a good state. The expected return of the good state is described by:

$$r_t = \mu_1 dt + \sigma_1 dZ_t \quad (15)$$

where $\mu_1 = 0.10$, $\sigma_1 = 0.10$, and Z_t is white noise. The expected return of the bad state is described by:

$$r_t = \mu_2 dt + \sigma_2 dZ_t \quad (16)$$

where $\mu_2 = -0.05$ and $\sigma_2 = 0.05$. If the asset price increased, participants received the highest profit if they use their entire endowment to purchase the asset. If the asset price decreased, participants received the lowest loss by purchasing none of the asset. This allows us to account for the fictive error occurring in the market process. We can define the fictive error as:

$$f^+ = (100\% * r_t^+) - (Allocation * r_t^+), f^- = (0\% * r_t^-) - (Allocation * r_t^-) \quad (17)$$

where r^+ is a positive return on the asset and r^- is a negative return on the asset, *Allocation* is the percentage of the participant's endowment that they used to purchase the risky asset, $Allocation * r_t^+$ is the experienced gain for the positive return, and $Allocation * r_t^-$ is the experienced loss for the negative return.

In both the human and algorithm conditions, the suggestion from the advisor was derived using the same Bayesian investment model. This allowed for a well-defined probability that the market was in a good state. Assume that r_t is a price change of the risky asset that the participant observed in period t . We then know that the probability that the market is currently in a good state is:

$$q_t = \Pr(s_t = good | r_t, r_{t-1}, \dots, r_2, r_1) \quad (18)$$

Therefore, we know:

⁵ Simulations were run to identify an optimal switch rate that balanced the predictability of market fluctuations with the desire for the algorithm to consistently outperform the individual investors.

$$q_t(q_{t-1}, r_t) \tag{19}$$

$$= \frac{\Pr(r_t | s_t = \text{good}) \Pr(s_t = \text{good} | q_{t-1})}{\Pr(r_t | s_t = \text{good}) \Pr(s_t = \text{good} | q_{t-1}) + \Pr(r_t | s_t = \text{bad}) \Pr(s_t = \text{bad} | q_{t-1})}$$

Further, we know the expected return in a good state from (10). We can define the distribution function of the expected return $\Pr(r_t | s_t = \text{good})$ as $f_{\text{good}}(r_t) = \frac{1}{\sigma_1 \sqrt{2\pi}} e^{-\frac{(r_t - \mu_1)^2}{2\sigma_1^2}}$. Similarly, in a bad state, the distribution function would be $f_{\text{bad}}(r_t) = \frac{1}{\sigma_2 \sqrt{2\pi}} e^{-\frac{(r_t - \mu_2)^2}{2\sigma_2^2}}$. Given the updated belief that the previous trial was in a good state q_{t-1} , $\Pr(s_t = \text{good} | q_{t-1}) = (1 - p)q_{t-1} + p(1 - q_{t-1})$ and $\Pr(s_t = \text{bad} | q_{t-1}) = pq_{t-1} + (1 - p)(1 - q_{t-1})$ where p is the state switch rate described above ($p = 0.35$). Therefore, we can rewrite (14) as following:

$$q_t(q_{t-1}, r_t) \tag{20}$$

$$= \frac{f_{\text{good}}(r_t)((1 - p)q_{t-1} + p(1 - q_{t-1}))}{f_{\text{good}}(r_t)((1 - p)q_{t-1} + p(1 - q_{t-1})) + f_{\text{bad}}(r_t)(pq_{t-1} + (1 - p)(1 - q_{t-1}))}$$

The algorithm then, derives $q_t(q_{t-1}, r_t)$ based on the previous round's return to determine the probability of this round being in a good state. We then derive the optimal amount of the endowment to use using the maximization of the utility function subject to the relationships between expected return and standard deviation of the risky asset as follows:

$$y^* = \frac{q_t \mu_1 + (1 - q_t) \mu_2}{0.02 A \sigma^2} \tag{21}$$

where A is a measure of risk tolerance for an individual. To simplify the experiment, rather than measure individual risk preferences in advance we used the average risk aversion level ($A=5.17$) from Holt and Laury (2002)) shown in (21).

Appendix B: Risk Disclaimers

Disclaimer for Betrayal Treatments (Verbiage in brackets refers to the human condition):

“In the past 10 rounds, you were given the option to use a *financial investment algorithm* [financial advisor, Brandon] to make your investment decisions. Historically, *the algorithm* [Brandon] has outperformed those who choose to invest themselves, where the majority of negative returns *it* [he] received were the result of random market volatility. However, *investment algorithms, like the one used in this experiment, are designed to earn revenue* [he is compensated] based on how frequently *it* [he] invests. Occasionally *the algorithm* [he] will intentionally over-invest even when *the algorithm* [he] is not confident that it is a good market. This incentive misalignment comes at the expense of increasing your risk and accounts for a small proportion of historical negative returns.”

Disclaimer for Error Treatments (Verbiage in brackets refers to the human condition)

“However, *investment algorithms, like the one used in this experiment*, [his recommendations] are not always perfect and [he] has occasionally made accidental errors that result in a negative return. These errors come at the expense of increasing your risk and account for a small proportion of historical negative returns.”

Appendix C: Comprehension Quiz

Each question had the following possible answers: True, False

Question	Answer
You are being asked to invest your research points in a market with TWO possible assets.	False – there is only ONE asset.
You can invest 0% of your research points in the asset.	True.
If the CURRENT market is good, then there is a higher chance of the NEXT market being bad.	False – If the current market is good, there is a higher chance of the next market being good.
1000 research points translates to \$1 at the end of the experiment.	False – 1000 research points translates to \$5 at the end of the experiment.
If you have any questions throughout the experiment, you should ask them through Zoom only.	True.

Appendix D: Advisor Interface

Admin Page

Round 1:

The suggested investment is 48%.

When available, please click next below if you would like to submit the suggestion.

Next

Appendix E: Exit Questions

Each question was answered using a Likert scale with the following responses: Strongly Agree, Agree, Neither Agree nor Disagree, Disagree.

Question
If I chose to invest with help, I was concerned about being misled.
If I chose to invest with help, I was concerned about my trust being violated.
If I chose to invest with help, I would feel betrayed if I received a negative return.
If I chose to invest with help, I trusted that a good investment decision would be made for me.
If I chose to invest with help, I felt like I made a mistake if I received a negative return.
If I chose to invest on my own, I felt like I made a mistake if I received a negative return.

Appendix F: Prior Advisor Performance with Interactions

VARIABLES	(1) Use
Betrayal Risk	-0.0673** (0.0314)
Error Risk	0.00280 (0.0339)
Algorithm	-0.0197 (0.0335)
Algorithm $\times$ Betrayal	0.0944** (0.0446)
Algorithm $\times$ Error	0.0361 (0.0461)
Prior Round Return	-0.000191*** (0.00004)
Prior Round Advisor Use	0.552*** (0.0220)
Prior Round Return $\times$ Prior Round Advisor Use	0.000553*** (0.00010)
Prior Round Return $\times$	-0.000269

Prior Round Advisor Use × Betrayal	(0.000227)
Prior Round Return ×	0.00007
Prior Round Advisor Use × Betrayal × Algorithm	(0.000260)
Prior Round Return ×	0.000118
Prior Round Advisor Use × Error	(0.000160)
Prior Round Return ×	-0.000256
Prior Round Advisor Use × Error × Algorithm	(0.000177)
Constant	0.217*** (0.0337)
Observations	8,250
Number of ID	275

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Appendix G: First Advisor Exposure

VARIABLES	(1: Negative) Use	(2: Positive) Use	(3: All) Use
Betrayal Risk	-0.165 (0.128)	-0.205 (0.125)	-0.165 (0.127)
Error Risk	0.131 (0.144)	-0.0827 (0.113)	0.131 (0.143)
Algorithm	0.105 (0.137)	-0.101 (0.115)	0.105 (0.136)
Algorithm × Betrayal	0.0823 (0.175)	0.208 (0.161)	0.0823 (0.174)
Algorithm × Error	-0.159 (0.185)	0.120 (0.153)	-0.159 (0.184)
Positive First Exposure			0.122 (0.136)
Algorithm × Positive First Exposure			-0.205 (0.177)
Betrayal × Positive First Exposure			-0.0399 (0.178)
Error × Positive First Exposure			-0.214 (0.182)
Algorithm × Positive First Exposure × Betrayal			0.126 (0.237)
Algorithm × Positive First Exposure × Error			0.280 (0.239)
Constant	0.445***	0.555***	0.400***

	(0.109)	(0.0997)	(0.106)
Observations	2,970	3,330	6,300
Number of ID	99	111	210

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Appendix H: Experiment 3 Vignettes

Context	Vignette
<i>Financial Investment</i>	Consider a financial advisor, Michael [<i>a financial algorithm, ProfitPulse</i>], who provides recommendations to their [<i>its</i>] clients. Michael [<i>ProfitPulse</i>] is incentivized with a yearly bonus [<i>through a set reward function</i>]. A portion of this bonus [<i>reward function</i>] is meant to prioritize investment recommendations that align with firm priorities. To maximize bonus [<i>reward function</i>], Michael [<i>ProfitPulse</i>] knowingly recommends an investment that exceeds his [<i>its</i>] client's risk tolerance, potentially jeopardizing their financial stability.
<i>Product Recommendations – Online Retail</i>	Consider a website administrator, Samantha [<i>a website management algorithm, ShopMatch</i>], who is employed [<i>which is utilized</i>] by an online retail store. One of her [<i>its</i>] responsibilities is to make product placement decisions for the store. She has been told [<i>It has been given data showing</i>] that there is a significant amount of remaining inventory for last season's products. Samantha [<i>ShopMatch</i>] chooses to prioritize these products in customer search results, leading to the purchasing of outdated or inferior products.
<i>Customer Support</i>	Consider a customer service representative, Steven [<i>customer service algorithm, ServiceSense</i>]. He is incentivized with quarterly evaluations [<i>It is incentivized through a set reward function</i>]. These evaluations are [<i>This reward function is</i>] partially based on the number of customer queries he [<i>it</i>] resolves each day. To maximize this number, Steven [<i>ServiceSense</i>] deliberately provides customers with incorrect or misleading information to close customer tickets more rapidly. This leaves customers with unresolved issues or incorrect solutions.
<i>Recruitment & Hiring</i>	Consider a recruiter, Emily [<i>recruitment algorithm, TalentFinder</i>], who [<i>which</i>] is responsible for selecting candidates for interviews at a company. Through her experience with the company [<i>its analysis of historical company data</i>], Emily [<i>TalentFinder</i>], has learned that individuals from a certain demographic group have a higher turnover rate. Therefore, she [<i>it</i>] intentionally rejects otherwise qualified individuals who come from that group.
<i>Education</i>	Consider a grader [<i>grading algorithm</i>] for a standardized test, Kevin [<i>ScoreStream</i>]. Kevin [<i>ScoreStream</i>] is incentivized by a performance evaluation [<i>through a reward function</i>] partially based on results of a survey sent to various college administrators that reflects the perceived rigor and high standards maintained by the assessment. Over time, Kevin

	[<i>ScoreStream</i>] has systematically awarded lower marks to students without regard to their objective performance on the exam.
<i>Automotive Manufacturing</i>	Consider a quality assurance specialist [<i>algorithm</i>], Heather [<i>QualiCheck</i>], who is employed [<i>which is utilized</i>] by a major car manufacturer and is involved in assessing vehicle safety. Heather [<i>QualiCheck</i>] is incentivized with a performance bonus [<i>through a reward function</i>] that is based on her ability [<i>that seeks to</i>] to minimize the operational costs she [<i>it</i>] incurs. Upon becoming aware of a critical flaw in the brake system of one of their popular car models, Heather [<i>QualiCheck</i>] fails to report the flaw to avoid the cost of a recall. This compromises the safety of the vehicle, leading to several accidents.
<i>Healthcare</i>	Consider a hospital system's director of operations, Mark [<i>healthcare operations algorithm, MediSync</i>]. Mark [<i>MediSync</i>] is responsible for allocating a limited-supply vaccine within a hospital. Despite the pressing need to prioritize front line workers at higher risk of disease exposure, Mark [<i>MediSync</i>] prioritizes age in his [<i>its</i>] decision, leading to a large portion of the vaccines being distributed to remote employees over in-person hospital employees. This decision compromises the health of the front line workers and their patients.
<i>Insurance</i>	Consider a claims representative, Carrie [<i>claims assessment algorithm, ClaimSwift</i>], employed [<i>utilized</i>] by an insurance company. Carrie [<i>ClaimSwift</i>] is incentivized by a performance evaluation [<i>through a reward function</i>] that considers how often her [<i>its</i>] decisions are appealed and overturned. Therefore, Carrie [<i>ClaimSwift</i>] tends to accept claims from individuals she [<i>it</i>] believes are likely to appeal if their claim was initially rejected. This practice leads to unfair claim approvals in which an individual has a higher likelihood of their claim being rejected if they are not likely to fight back.
<i>Social Media Content Moderation</i>	Consider a social media content moderator, Drew [<i>social media content moderation algorithm, SafeStream</i>]. Drew [<i>SafeStream</i>] is responsible for enforcing the platform's community guidelines. Motivated by personal political beliefs [<i>political ideologies learned from historical data</i>], Drew [<i>SafeStream</i>] inconsistently applies these guidelines when reviewing content. This behavior leads to an imbalanced representation of opinions on the platform.
<i>Delivery Operations</i>	Consider a delivery service manager, Jessica [<i>delivery service algorithm, RouteSmart</i>], responsible for overseeing the routing and dispatch of packages. Jessica is evaluated [<i>RouteSmart is incentivized through a reward function</i>] based on the efficiency of her [<i>its</i>] operations, with a yearly bonus partially tied to minimizing operational costs [<i>and the minimizing of operational costs</i>]. To maximize this bonus [<i>reward function</i>], Jessica [<i>RouteSmart</i>] intentionally redirects high-priority packages to a slower, less costly route. This decision, while decreasing immediate operational expenses, causes significant delays in the delivery of critical packages, adversely affecting customers who rely on timely deliveries for their essential needs and who paid for expedited shipping.

Appendix I: Experiment 3 Questions

Question	Response
<i>The impacted party has filed a lawsuit requesting monetary damages. Does the impacted party deserve compensation for the harm they endured?</i>	<i>Binary: Yes, No.</i>
<i>The average award in this type of lawsuit is indicated on the slider bar below. If fault is proven, how much should the impacted party be awarded?</i>	<i>Continuous slider with labels: Much less than average, Slightly less than average, Average, Slightly more than average, Much more than average.</i>
<i>If you were the plaintiff, you would be unlikely to use [context specific service] in the future.</i>	<i>Likert scale: Strongly disagree, Disagree, Somewhat disagree, Neither agree nor disagree, Somewhat agree, Agree, Strongly agree.</i>
<i>If you were the plaintiff, you would feel betrayed.</i>	<i>Likert scale: Strongly disagree, Disagree, Somewhat disagree, Neither agree nor disagree, Somewhat agree, Agree, Strongly agree.</i>
<i>If you were the plaintiff, you would prefer that [human/algorithm] be replaced by a [human/algorithm].</i>	<i>Likert scale: Strongly disagree, Disagree, Somewhat disagree, Neither agree nor disagree, Somewhat agree, Agree, Strongly agree.</i>
<i>What percent of the blame would you attribute to [name of actor]?</i>	<i>Continuous: 0-100%.</i>